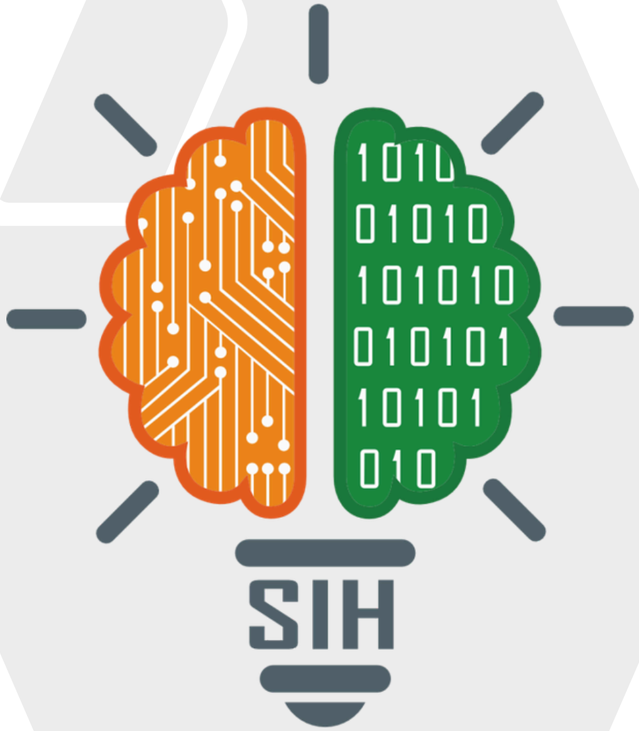
SMART INDIA HACKATHON 2026

ResQVision



- **Problem Statement ID – 26177**
- **Problem Statement Title-** A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- **Theme-** Robotics and Drones
- **PS Category-** Hardware
- **Team ID-**151224
- **Team Name (Registered on portal)-** .HackSmiths

.HackSmiths

ResQVision

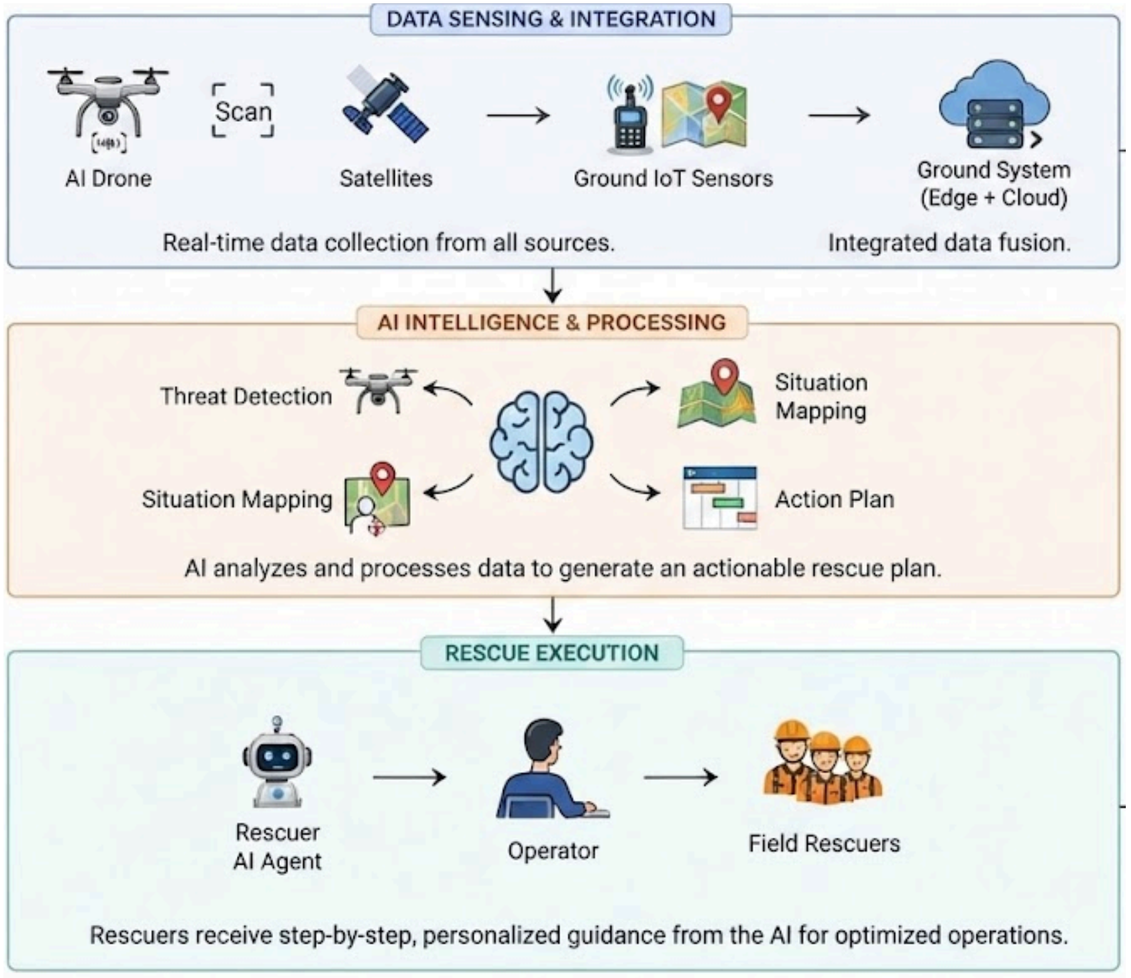
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OBJECTIVE

Make disaster rescue more effective and time-efficient through AI-assisted operations.

Proposed Solution

1. **AI-Powered Drone :** Detect threats and process visuals instantly, assisting recuse teams with intelligent navigation .
2. **Multi-source intelligence :** Combining data from external sources (like satellites, maps, or ground sensors) in addition to the drone's view.
3. **Structured incident intelligence :** Every threat with critical data like where, when, and how server it is, ensuring no detail is missed, consequently system performance is enhanced .
4. **Adaptive recuse plannig :** The system dynamically updates the rescue strategy in real time as victim locations, hazards, team numbers, and road access shift.
5. **Rescuer AI Agent :** Converts complex, city-wide emergency data into a personalized, step-by-step field guide for each individual responder.



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TECHNICAL APPROACH

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Technology Stack

🧠 **AI / ML:** PyTorch, OpenCV, OpenAI API, Gemini API, Kaggle Datasets

⚙️ **Backend & Data_:** FastAPI, SQLAlchemy, Pydantic, PostgreSQL, Celery, Redis, REST APIs

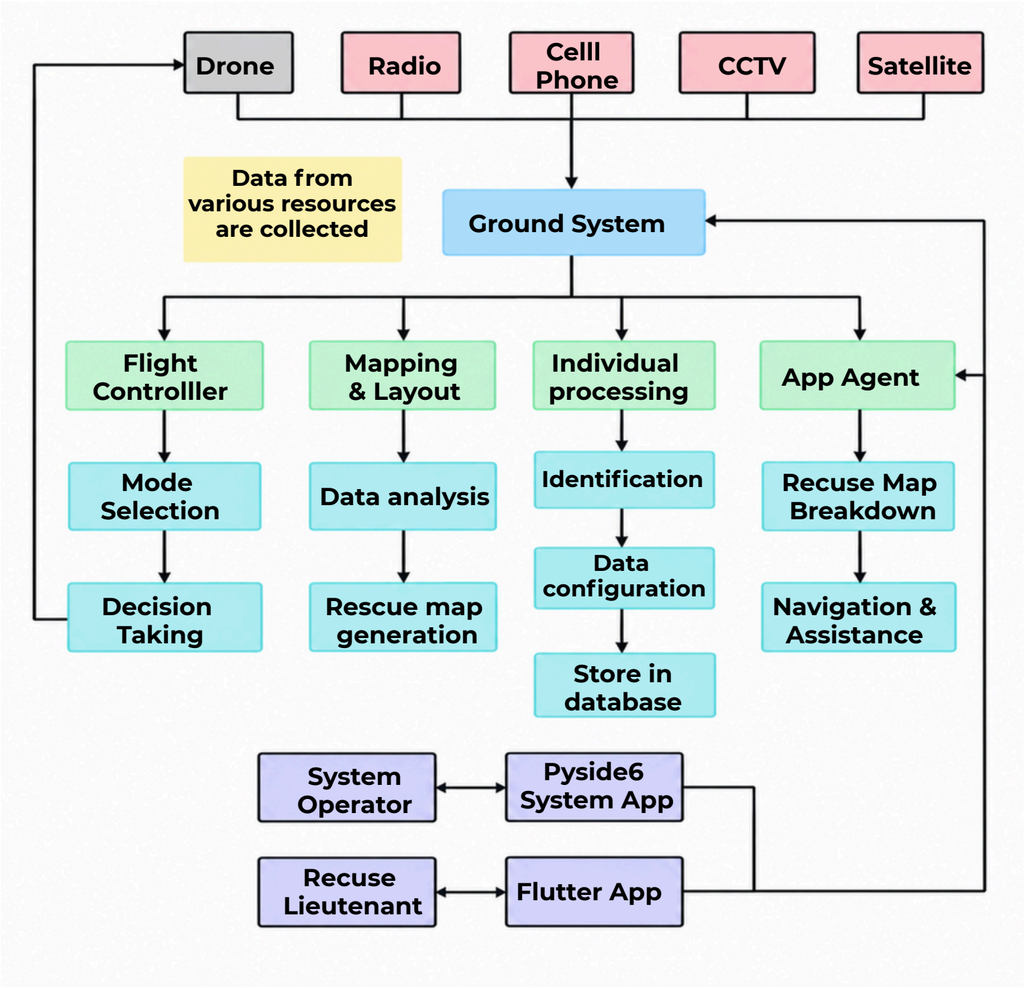
🖥️ **Ground System:** PySide6, Geospatial / Mapping Libraries

📱 **Mobile Application:** Flutter, Riverpod, Sqflite

📡 **Connectivity** -ExpressLRS (ELRS), 5.8 GHz FPV, WebSocket, WebRTC, Wi-Fi / Internet, Cloudflare Tunnel

🔐 **Security:** HTTPS / TLS, JWT Authentication, Encrypted Data Storage

🛠️ **Development & Infrastructure:** Git, Linux, Docker, Vercel, Cloudflare



Embedded Systems

Flight Control — DYS FC722 F7
Positioning — GPS, IMU(built-in)
Perception — RGB Camera + Ultrasonic + Barometer(built-in)
Control Link — RP1 ELRS
Video Link — 5.8 GHz FPV

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FEASIBILITY AND VIABILITY

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⚙️ Technical Feasibility

- **Ground-Centric Processing**: AI processing is handled by the ground system to reduce drone computing and battery load.
- **Multi-Source Integration :** Designe to combine drone, maps, CCTV, Satellite, and authorized field data where available.
- **Human-In-Loop :** AI provides detection, route and hazard recommendations while critical decisions remain with responders.

💼 Commercial Viability

- **Cost-Efficient Scalability**: Employs commercial off-the-shelf hardware (COTS) instead of building expensive military-grade drones.
- **Low Maintainability Costs**: Features swappable subsystems so damaged drone parts can be replaced instantly in the field.
- **Software-Defined Upgrades**: Allows AI models to be updated remotely over the air without changing the physical drone platform.

🎯 Strategic Use Cases

- **Flood & Landslide** : Detect stranded people, identify blocked routes and support rescue planning.
- **Structural Disaster Response**: Survey affected areas, identify possible survivors and map hazardous zones.
- **Disaster Assessment :** Combine aerial and field information into a continuously update incident map.

⚠️ System Challenges & Mitigations

- **Signal Loss & Failures:** Semi-autonomous failsafe attempts communication recovery and then returns to the predefined safe location/base.
- **Network Failure :** System supports degraded/local communication; future mesh networking can improve connectivity during infrastructure outages.
- **Environmental Constraints:** Visible and Thermal imaging can improve detection in smoke, darkness and reduced visibility.

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IMPACT AND BENEFITS

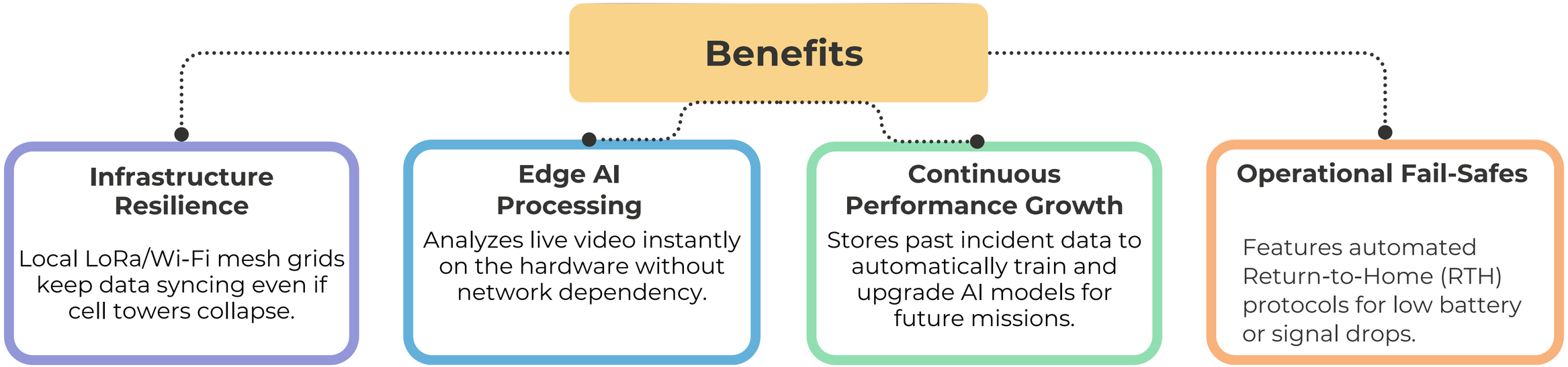
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Rescue Teams (Operational Safety)

- **Personalized AI Copilots:** Converts complex disaster maps into simple, individual step-by-step checklists.
- **Live Hazard Avoidance:** Dynamically reroutes ground teams away from spreading fires or collapsing rubble.
- **Airspace & Signal Safety:** Uses human-pilot overrides to prevent collisions with manned recuse helicoptes.

Survivors (Life-Saving Speed)

- **Slashes Detection Time:** Ground-centric AI spots human shapes instantly, eliminating long manual footage reviews.
- **Zero Forgetten Victims:** Permanently logs every survivor's data until their rescue stauts is marked "Secured".



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RESEARCH AND REFERENCES

CASE STUDIES

- **UAV Victim Detection-**deep learning used to detect disaster victims from aerial imagery https://www.mdpi.com/2072-4292/14/13/2977?utm_source=chatgpt.com
- **NDMA SACHET —** geo-targeted, multi-source disaster alerts using geo-intelligence.

RESEARCH PAPERS & REPORTS

- **Daud et al. (2022) —** Applications of Drone in Disaster Management: A Scoping Review
- **Zhang et al. (2022) —** Training a Disaster Victim Detection Network for UAV Search and Rescue

TECHNICAL REFERENCES

- **PyTorch** — deep-learning model development
- **WebRTC —** real-time video communication
- **YOLO / Ultralytics —** real-time object detection

Capability	Capability	ResQVision
Aerial victim detection	✔	✔
Hazard detection	✔	✔
Drone Autonomy	✔	✔
Geo-tagged mapping	✔	✔
Multi-source Intelligence	Limited	✔
Structured Incident State	Fragmented	✔
Resource aware	Limited	✔
Dynamic Recuse planning	Limited	✔
Rescurer specific AI-agent	Not Central	✔